



Coimisiún na Scrúduithe Stáit
State Examinations Commission

Leaving Certificate 2025
Deferred Examinations

Marking Scheme

Mathematics

Higher Level

Note to teachers and students on the marking schemes for the deferred examinations

Marking schemes published by the State Examinations Commission are not intended to be standalone documents. They are an essential resource for examiners who receive training in the correct interpretation and application of the scheme. However, it should be noted that the marking schemes for the deferred examinations may not necessarily be as detailed as the corresponding marking schemes for the main sitting of an examination, which serve to ensure consistency across a large team of examiners.

Marking schemes are working documents. While a draft marking scheme is prepared in advance of the examination, the scheme is not finalised until examiners have applied it to candidates' work and the feedback from examiners has been collated and considered in light of the full range of responses of candidates, the overall level of difficulty of the examination, and the need to maintain consistency in standards between the main sitting and the deferred sitting and from year to year. In the case of the deferred examinations, this means that the level of detail may vary by question, as the marking scheme will only have been finalised for the questions attempted by the candidates who sat these examinations.

In the case of marking schemes that include model solutions or answers, it should be noted that these are not intended to be exhaustive. Variations and alternatives may also be acceptable. Examiners must consider all answers on their merits, and will have consulted with a senior examiner when in doubt.

Future Marking Schemes

Assumptions about future marking schemes on the basis of past schemes (whether for the main examinations or the deferred examinations) should be avoided. While the underlying assessment principles remain the same, the details of the marking of a particular type of question may change in the context of the contribution of that question to the overall examination concerned. Accordingly, aspects of the structure, detail and application of the marking scheme for a particular examination will not necessarily be the same for the deferred sitting as for the main sitting or from one year to the next.



Coimisiún na Scrúduithe Stáit
State Examinations Commission

Leaving Certificate Examination 2025

Deferred Examination 2025

Mathematics

Higher Level

Paper 2

Marking scheme

300 marks

Structure of the marking scheme – Paper 2

Candidate responses are marked according to different scales, depending on the types of response anticipated. Scales labelled **A** divide candidate responses into two categories (correct and incorrect), scales labelled **B** divide responses into three categories (correct, partially correct, and incorrect), and so on. The scales and the marks that they generate are summarised in this table:

Scale label	A	B	C	D
Number of categories	2	3	4	5
5-mark scales		0, 2, 5	0, 2, 3, 5	
10-mark scales		0, 5, 10	0, 4, 7, 10	0, 3, 5, 7, 10
15-mark scale			0, 5, 10, 15	0, 4, 7, 11, 15

A general descriptor of each point on each scale is given below. More specific directions in relation to interpreting the scales in the context of each question are given in the scheme.

Marking scales – level descriptors

A-scales (two categories)

- incorrect response
- correct response

B-scales (three categories)

- response of no substantial merit
- partially correct response
- correct response

C-scales (four categories)

- response of no substantial merit
- response with some merit
- almost correct response
- correct response

D-scales (five categories)

- response of no substantial merit
- response with some merit
- response about half-right
- almost correct response
- correct response

General notes on the marking – Paper 2

In certain cases, typically involving incorrect rounding, omission of units, a misreading that does not oversimplify the work, or an arithmetical error that does not oversimplify the work, a mark that is one mark below the full-credit mark may also be awarded. Such cases are denoted with a * and this level of credit is referred to as *Full Credit -1*. Thus, for example, in Scale 10C, *Full Credit -1* of 9 marks may be awarded. The only marks that may be awarded for a question are those on the scales above, or *Full Credit -1*.

Instructions regarding penalties for omitted or incorrect units are given in the scheme for each question to which they apply. A penalty for rounding is applied once per unit of marking (i.e., if parts (a) and (b) are marked as a single unit, a penalty for rounding is only applied once for (a) and (b) combined).

In general, an answer without sufficient supporting work is awarded the highest level of credit on the scale below *Full Credit* (typically *Partial Credit* or *High Partial Credit*, as appropriate).

In general, accept a candidate's work in one part of a question for use in subsequent parts of the question, unless this oversimplifies the work involved.

Steps

Where steps are listed in the Marking Notes, unless otherwise specified, it is to be taken that they can be independently correct / incorrect – that is, in a candidate's solution, step n can be considered correct even if previous step(s) have not been correctly presented, as long as the work done to arrive at step n from the previous step(s) has not been oversimplified. It is specifically noted where this does not hold. Note also that these steps may not need to be presented in the order specified in the Marking Notes.

Errors

Where a question is **not** marked using steps, if a candidate has a single error, they are generally awarded one level of credit below that which they would otherwise have been awarded. Similarly, where they have two errors, they are generally awarded two levels of credit below that which they would otherwise have been awarded. (If they present sufficient work for *Low Partial Credit*, they will be awarded this at a minimum, regardless of the number of errors.) For example, on a C-scale:

- *High Partial Credit*: One error, otherwise fully correct (or fully correct with a *)
- *Low Partial Credit*: Two errors, otherwise fully correct (or fully correct with a *)

Where a question **is** marked using steps, this does not apply. Instead, an error in a step means that the step has not been completed correctly; this does not affect the completion of other steps (unless it oversimplifies the work). So if a candidate has multiple errors on a single step, they could still be awarded up to *High Partial Credit*, depending on the marking scheme.

Slips and *

Where a candidate has a single * on their solution, this is ignored in the awarding of credit unless they would otherwise have *Full Credit*. Where a candidate has multiple *s, this is generally treated as an error.

Multiple answers

Where the solution requires substantial work, mark all separate attempts and award the marks for the best one, regardless of crossing out.

Where a solution requires selection from the question:

- If a candidate has crossed out answer(s), ignore the crossed-out answers
- If a candidate has multiple answers that are **not** crossed out, award the lowest mark associated with these answers (generally, this will be considered incorrect)

Square brackets

Where something is contained in square brackets in the model solution, it is **not** required for *Full Credit*.

Work of merit

Where the scheme indicates “work of merit”, examples are given to exemplify the standard of work required to be considered work of merit in that particular question.

Palette of annotations available to examiners – Paper 2

Symbol	Name	Meaning in the body of the work	Meaning when used in the right margin
	Tick	Work of relevance	The work presented in the body of the script merits full credit
	Cross	Incorrect work (distinct from an error)	The work presented in the body of the script merits 0 credit
	Star	Rounding / Unit / Arithmetic error / Misreading	
	Horizontal wavy	Error	
	P		The work presented in the body of the script merits <i>Partial Credit</i>
	L		The work presented in the body of the script merits <i>Low Partial Credit</i>
	M		The work presented in the body of the script merits <i>Mid Partial Credit</i>
	H		The work presented in the body of the script merits <i>High Partial Credit</i>
	F star		The work presented in the body of the script merits <i>Full Credit – 1</i>
	Left Bracket		Another version of this solution is presented elsewhere and it merits equal or higher credit
	Vertical wavy	No work on this page / portion of this page	
	Oversimplify	The candidate has oversimplified the work	
	Work of merit	The candidate has produced work of merit (in line with that defined in the scheme)	
	Stops early	The candidate has stopped early in this part	

Note: Where work of substance is presented in the body of the script, the annotation on the right margin should reflect a combination of annotations in the work.

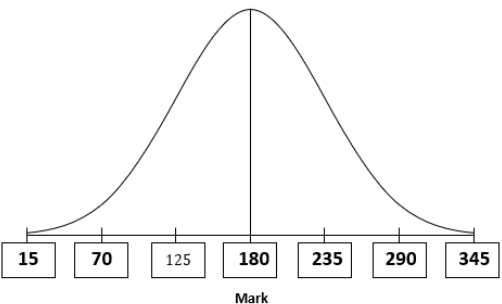
In a **C scale** that is **not** marked using steps, where * and  and  appear in the body of the work, then  should be placed in the right margin.

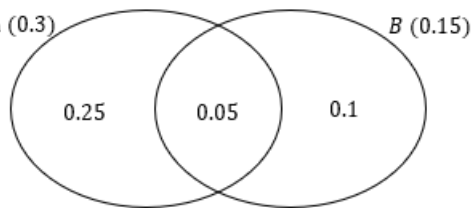
In the case of a **D scale** with the same annotations,  should be placed in the right margin.

Model Solutions & Marking Notes – Paper 1

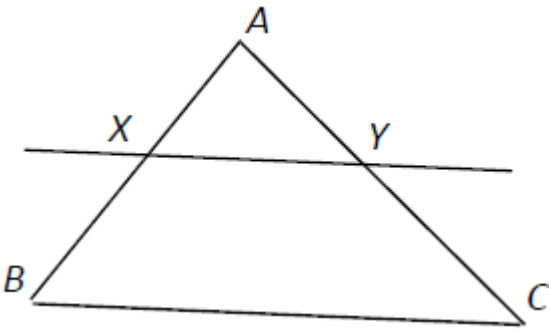
Note: The model solutions for each question are not intended to be exhaustive – there may be other correct solutions. Any Examiner unsure of the validity of the approach adopted by a particular candidate to a particular question should contact his / her Advising Examiner.

Q1	Model Solution – 30 Marks	Marking Notes
(a)(i)	$m_1 = \frac{1}{2}$ $m_2 = -1$	Scale 10C (0, 4, 7, 10) <i>Low Partial Credit</i> - Some effort at finding a slope <i>High Partial Credit</i> - One slope found
(a)(i)	$\tan \theta = \frac{\frac{1}{2} - (-1)}{1 + \frac{1}{2}(-1)} = 3$ $\tan^{-1} 3 = 71.56 \dots$ $= 71.6^\circ$	Scale 10C (0, 4, 7, 10) <i>Low Partial Credit</i> - Some correct substitution into formula <i>High Partial Credit</i> - Fully correct substitution into formula <i>Full Credit -1</i> - Apply a * for no or incorrect rounding - Apply a * for no or incorrect units
(b)	$y = -x$ $\frac{ x + y }{\sqrt{1^2 + 1^2}}$ $= \frac{ 4 + 2 }{\sqrt{2}}$ $= 3\sqrt{2}$	Scale 10C (0, 4, 7, 10) <i>Low Partial Credit</i> - Any correct substitution into formula - Finds the equation of the line AC <i>High Partial Credit</i> - Fully correct substitution into formula $\frac{ 1(4) + 1(2) }{\sqrt{1^2 + 1^2}}$ <i>Full Credit -1</i> - Apply a * for incorrect format

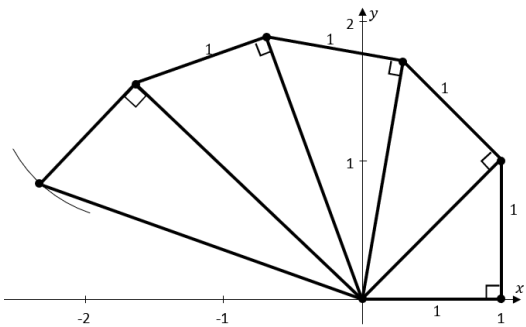
Q2	Model Solution – 30 Marks	Marking Notes
(a) (i) (ii)	(i)  (ii) $68 + 13.5 = 81.5\%$ $30\,000 \times 0.815 = 24\,450$	Scale 15D (0, 4, 7, 11, 15) <i>Low Partial Credit:</i> • Work of merit, for example, one correct entry in (i) • Work of merit, for example, a relevant percentage from the empirical rule in (ii) <i>Mid Partial Credit:</i> • Work of merit in (i) and (ii) • One part correct <i>High Partial Credit:</i> • One part correct and work of merit in the other part
(b) (i)	$0\left(\frac{3}{5}\right) + 100\left(\frac{1}{20}\right) + 10\left(\frac{1}{4}\right) + 25\left(\frac{1}{10}\right)$ $= 10$ $10 + 0.5$ $= 10.50$	Scale 10C (0, 4, 7, 10) <i>Low Partial Credit:</i> • A correct term formulated <i>High Partial Credit:</i> • All terms formulated
(b) (ii)	$\frac{1}{4}$ The result of the previous spin doesn't influence the next spin	Scale 5B (0, 2, 5) <i>Partial Credit:</i> • Correct answer no justification

Q3	Model Solution – 30 Marks	Marking Notes
(a) (i) (ii)	(i)  $P(A \setminus B) = P(A) - P(A \cap B)$ $P(A \setminus B) = 0.25$ (ii) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ $0.3 + 0.15 - 0.05$ $= 0.4$ OR $0.3 + 0.1 = 0.4$	Scale 10D (0, 3, 5, 7, 10) <i>Low Partial Credit:</i> • Work of merit, for example, some relevant subtraction in (i) • Work of merit, for example, a relevant Venn diagram drawn in (i) or (ii) <i>Mid Partial Credit:</i> • Work of merit in (i) and (ii) • One part correct <i>High Partial Credit:</i> • One part correct and work of merit in the other part
(b) (i) (ii)	(i) $x + 0.3$ (ii) 0	Scale 10C (0, 4, 7, 10) Accept correct answers without work <i>Low Partial Credit:</i> • Work of merit in (i) or (ii) <i>High Partial Credit:</i> • One part correct
(c)	$\frac{0.08}{x} = \frac{1}{3}$ $x = 0.24$	Scale 10C (0, 4, 7, 10) <i>Low Partial Credit:</i> • Relevant work on a Venn diagram <i>High Partial Credit:</i> • $\frac{0.08}{x}$

Q4	Model Solution – 30 Marks	Marking Notes
(a) (b)	(a) Reference angle = 45° or $\frac{\pi}{4}$ $360 - 45 = 315[^\circ]$ $2\pi - \frac{\pi}{4} = \frac{7\pi}{4}$ (b) $\cos\left(-\frac{\pi}{6}\right) - \sin\left(-\frac{\pi}{6}\right)$ $\frac{1}{2} - \left(-\frac{\sqrt{3}}{2}\right)$ 1.3660	Scale 10D (0, 3, 5, 7, 10) <i>Low Partial Credit:</i> • Work of merit, for example, reference angle found in (i) • Work of merit, for example, some correct substitution in (ii) <i>Mid Partial Credit:</i> • Work of merit in (i) and (ii) • One part correct <i>High Partial Credit:</i> • One part correct and work of merit in the other part <i>Full Credit -1</i> • Apply a * for incorrect format
(c)	$\sin 2x = \frac{\sqrt{3}}{2}$ $2x = 60$ $x = 30[^\circ]$ $2x = 120$ $x = 60[^\circ]$	Scale 10C (0, 4, 7, 10) <i>Low Partial Credit:</i> • Work of merit, for example, isolating $\sin 2x$ <i>High Partial Credit:</i> • One correct value for x
(d)	$A = 30$ $B = 250$ $C = \frac{2\pi}{3}$	Scale 10C (0, 4, 7, 10) <i>Low Partial Credit:</i> • Work of merit, for example, some understanding of the period or range • One value correct <i>High Partial Credit:</i> • Two values correct

Q5	Model Solution – 30 Marks	Marking Notes
(a)	<i>Diagram:</i>  Given: A triangle ABC and a line XY parallel to BC which cuts AB in the ratio $s : t$ where $s, t \in \mathbb{N}$. To Prove: $[AY] : [YC] = s : t$ Construction: Divide $[AB]$ into $s + t$ equal parts, s of them lying along $[AX]$ and t of them lying along $[XB]$. Through each point of division draw a line parallel to $[BC]$ Proof: By a previous theorem the parallel lines cut off segments of equal length along $[AC]$. Therefore $[AC]$ is divided into $s + t$ equal parts with s of them forming $[AY]$ and t of them forming $[YC]$. Let k be the length of one segment on $[AC]$. $[AY] : [YC] = ks : kt = s : t$	Scale 15D (0, 4, 7, 11, 15) <i>Low Partial Credit:</i> - Relevant diagram drawn <i>Mid Partial Credit:</i> - Construction clearly indicated <i>High Partial Credit:</i> - Proof missing 1 relevant step
(b)	Let $a = 8$ $\frac{4}{7} = \frac{8}{8 + d}$ $8 + d = 14$ $d = 6$ Or similar	Scale 15C (0, 5, 10, 15) <i>Low Partial Credit:</i> - Work of merit, for example, a relevant ratio found <i>High Partial Credit:</i> - One variable written in terms of the other

Q6	Model Solution – 30 Marks	Marking Notes
(a) (i)	Centre = $(2, -1)$ $r = \sqrt{2^2 + -1^2 + 20} = 5$	Scale 5C (0, 2, 3, 5) <i>Low Partial Credit:</i> • Work towards finding the centre or radius <i>High Partial Credit:</i> • Centre or radius correct
(a) (ii)	Centre = $(-14, 11)$, radius = 15 $ c_1c_2 = \sqrt{(-14 - 2)^2 + (11 - (-1))^2}$ $ c_1c_2 = 20$ $r_1 + r_2 = 5 + 15 = 20$ $ c_1c_2 = r_1 + r_2$	Scale 10D (0, 3, 5, 7, 10) <i>Low Partial Credit:</i> • Radius and/or centre of circle t found • Some correct substitution into distance formula • Some understanding of circles touching externally <i>Mid Partial Credit:</i> • $r_1 + r_2$ found <i>High Partial Credit:</i> • c_1c_2 found <i>Full Credit -1:</i> • Incorrect or no conclusion
(b) (i)	$h^2 + h^2 = 2^2$ $h = \sqrt{2}$ OR $\sin 45 = \frac{h}{2}$ $h = \frac{2}{\sqrt{2}} = \sqrt{2}$	Scale 5B (0, 2, 5) <i>Partial Credit:</i> • Work of merit, for example, some relevant work towards finding the value of h or relevant work on the diagram <i>Full Credit -1</i> • Apply a * for incorrect format
(b) (ii)	$l = 2 + \sqrt{2}$ $l^2 = (2 + \sqrt{2})^2$ $l^2 = 6 + 4\sqrt{2}$	Scale 10C (0, 4, 7, 10) <i>Low Partial Credit:</i> • Some relevant use of their value of h or some use of area of a square <i>High Partial Credit:</i> • Length of the side of the square found

Q7	Model Solution – 50 Marks	Marking Notes
(a)		Scale 10C (0, 4, 7, 10) <i>Low Partial Credit:</i> 90° angle correct or 1 unit correct <i>High Partial Credit:</i> 90° angle correct and 1 unit correct
(b) (i)	$\tan A = 1 \Rightarrow A = 45^\circ$ $\tan B = \frac{1}{\sqrt{2}} \Rightarrow B = 35.26$ $A + B = 80.26[^\circ]$	Scale 10C (0, 4, 7, 10) <i>Low Partial Credit:</i> 1 correct angle <i>High Partial Credit:</i> 2 correct angles
(b) (ii)	$\sqrt{3} \sin 80.26$ $= 1.707 \dots$ $= 1.71$	Scale 5C (0, 2, 3, 5) <i>Low Partial Credit:</i> - Work of merit, for example, use of trigonometry in co-ordinates or relevant work on the diagram <i>High Partial Credit:</i> - Finds the x co-ordinate
(c)	No The first triangle has two sides equal to one, the hypotenuse is always increasing while one side always equals one	Scale 5B (0, 2, 5) <i>Partial Credit:</i> Correct answer (No) given
(d)	$\sqrt{1025}$ $1025 - 1 = 1024$	Scale 5B (0, 2, 5) <i>Partial Credit:</i> $\sqrt{1025}$ found

Q7	Model Solution – 50 Marks	Marking Notes
(e)	$\sin(X + Y) = \sin X \cos Y + \cos X \sin Y$ $\sin X = \frac{1}{\sqrt{26}} \quad \cos X = \frac{5}{\sqrt{26}}$ $\sin Y = \frac{1}{\sqrt{27}} \quad \cos Y = \frac{\sqrt{26}}{\sqrt{27}}$ $\sin(X + Y) = \frac{1}{\sqrt{26}} \cdot \frac{\sqrt{26}}{\sqrt{27}} + \frac{5}{\sqrt{26}} \frac{1}{\sqrt{27}}$ $= \frac{5 + \sqrt{26}}{\sqrt{702}}$	Scale 15D (0, 4, 7, 11, 15) There are four steps involved in the solution: Step 1. Finds $\sin X$ and $\cos X$ Step 2. Finds $\sin Y$ and $\cos Y$ Step 3. Substitutes into the correct trigonometric formula Step 4. Finishes in correct form <i>Low Partial Credit:</i> • Work of merit, for example, $\sqrt{25}$ found or some relevant work with trigonometric ratios <i>Mid Partial Credit:</i> • Finds two of $\frac{1}{\sqrt{26}}, \frac{5}{\sqrt{26}}, \frac{1}{\sqrt{27}}, \frac{\sqrt{26}}{\sqrt{27}}$ • 1 step correct <i>High Partial Credit:</i> • 3 steps correct <i>Full Credit -1:</i> • Apply a * for incorrect format

Q8	Model Solution – 50 Marks	Marking Notes
(a) (i)	$\frac{x}{\sin 36} = \frac{73}{\sin 81}$ $x = 43.443 \dots$ $x = 43.4 \text{ m}$	Scale 10C (0, 4, 7, 10) <i>Low Partial Credit:</i> - Work of merit, for example, some correct substitution into the sine rule formula <i>High Partial Credit:</i> - Fully substituted sine rule formula
(a) (ii)	$73^2 = 58^2 + 41^2 - 2(58)(41) \cos \theta$ $4756 \cos \theta = -284$ $\theta = \cos^{-1} \left(-\frac{284}{4756} \right)$ $\theta = 93.423 \dots$ $\theta = 93.4^\circ$	Scale 10C (0, 4, 7, 10) <i>Low Partial Credit:</i> - Work of merit, for example, some correct substitution into the cosine rule formula <i>High Partial Credit:</i> - Fully substituted cosine rule formula <i>Full Credit -1</i> - Apply a * for no or incorrect units
(a) (iii)	$\text{Area } \triangle ABC = \frac{1}{2}(73)(43.4) \sin 63$ $= 1411.44 \dots$ $\text{Area } \triangle ACD = \frac{1}{2}(58)(41) \sin 93.4$ $= 1186.907 \dots$ $\text{Total area } ABCD = 2598.35 \dots$ $2598.35 \dots \times 0.2$ $= 519.67 \dots$ $= 520 \text{ [m}^3\text{]}$	Scale 10D (0, 3, 5, 7, 10) There are four steps involved in the solution: Step 1. Finds the area of $\triangle ABC$ Step 2. Finds the area of $\triangle ACD$ Step 3. Finds total area Step 4. Finds the volume <i>Low Partial Credit:</i> - Work of merit, for example, uses their answers from (i) or (ii) <i>Mid Partial Credit:</i> 2 steps correct <i>High Partial Credit:</i> 3 steps correct
(b) (i)	$20 \left(\frac{3\sqrt{3}}{2} L^2 \right) + 12 \left(\frac{L^2}{4} \sqrt{5(5 + 2\sqrt{5})} \right)$	Scale 5B (0, 2, 5) <i>Partial Credit:</i> Total area of hexagons or pentagons found in terms of L

Q8	Model Solution – 50 Marks	Marking Notes
(b) (ii)	S.A. of sphere $4\pi(11)^2$ $= 484\pi$ $L^2 \left(30\sqrt{3} + 3\sqrt{25 + 10\sqrt{5}} \right) = 484\pi$ $L^2 = 20.941 \dots$ $L = 4.576 \dots$ $L = 4.6 \text{ cm}$	Scale 10D (0, 3, 5, 7, 10) There are three steps involved in the solution: Step 1. Finds the surface area of the sphere Step 2. Equates surface area of sphere to answer from (i) Step 3. Finishes <i>Low Partial Credit:</i> • Work of merit, for example, radius found or some substitution into surface area of a sphere formula or uses their answer from (i) <i>Mid Partial Credit:</i> • 1 step correct <i>High Partial Credit:</i> • 2 steps correct
(c)	$l^2 = n x^2$ $n = \frac{l^2}{x^2}$ $\frac{l}{x} = \sqrt{n}$ $l^3 = k x^3$ $k = \frac{l^3}{x^3}$ $\frac{l}{x} = \sqrt[3]{k}$ $\sqrt[3]{k} = \sqrt{n}$ $k = n^{\frac{3}{2}}$	Scale 5D (0, 2, 3, 4, 5) There are three steps involved in the solution: Step 1. Surface area equation Step 2. Volume equation Step 3. Finishes <i>Low Partial Credit:</i> • Work of merit, for example, some substitution into surface area or volume of a cube <i>Mid Partial Credit:</i> • 1 step correct <i>High Partial Credit:</i> • 2 steps correct

Q9	Model Solution – 50 Marks	Marking Notes																
(a) (i)	<table><tr><td></td><td>Voted Alayah</td><td>Voted David</td><td>Total</td></tr><tr><td>Rural Voter</td><td>10%</td><td>8</td><td>18%</td></tr><tr><td>Urban Voter</td><td>51.8</td><td>30.2</td><td>82</td></tr><tr><td>Total</td><td>61.8%</td><td>38.2</td><td>100%</td></tr></table>		Voted Alayah	Voted David	Total	Rural Voter	10%	8	18%	Urban Voter	51.8	30.2	82	Total	61.8%	38.2	100%	Scale 10C (0, 4, 7, 10) <i>Low Partial Credit:</i> 1 correct new entry <i>High Partial Credit:</i> 3 correct new entries
	Voted Alayah	Voted David	Total															
Rural Voter	10%	8	18%															
Urban Voter	51.8	30.2	82															
Total	61.8%	38.2	100%															
(a) (ii)	$P(D) + P(A) - P(D \cap A)$ $= 38.2 + 82 - 30.2$ $= 90\%$	Scale 5B (0, 2, 5) <i>Low Partial Credit:</i> $P(D)$ or $P(A)$																
(a) (iii)	$P(\text{Urban}/\text{Alayah})$ $= \frac{51.8}{61.8}$ $= 0.8382$	Scale 5C (0, 2, 3, 5) <i>Low Partial Credit:</i> 51.8 or 61.8 <i>High Partial Credit</i> $\frac{51.8}{61.8}$																
(a) (iv)	$P(P \cap Q) = P(P).P(Q)$ $(P \cap Q) = 0.518$ $P(P).P(Q) = (0.618)(0.82)$ $= 0.50676$ $0.518 \neq 0.50676$ $\Rightarrow \text{Not independent}$	Scale 5D (0, 2, 3, 4, 5) <i>Low Partial Credit:</i> - Effort at independent formula <i>Mid Partial Credit:</i> $P(P \cap Q) = 0.518$ $P(P).P(Q) = 0.50676$ <i>High Partial Credit:</i> $P(P \cap Q) = 0.518$ and $P(P).P(Q) = 0.50676$																

Q9	Model Solution – 50 Marks	Marking Notes
(b) (i) (ii)	(i) $9!$ $= 362\,880$ (ii) $7! \times 3!$ $30\,240$	Scale 10D (0, 3, 5, 7, 10) <i>Low Partial Credit:</i> • Work of merit, for example, some relevant multiplication in (i) • Work of merit, for example, some understanding of arrangements in (ii) <i>Mid Partial Credit:</i> • Work of merit in (i) and (ii) • One part correct <i>High Partial Credit:</i> • One part correct and work of merit in the other part
(c) (i)	0.65^6 0.0754	Scale 10B (0, 5, 10) <i>Partial Credit:</i> • Work of merit, 0.65 written
(c) (ii)	$P(6) = 0.35^6 = 0.001838 \dots$ $P(5) = \binom{6}{5} (0.35^5)(0.65) = 0.02048 \dots$ $P(4) = \binom{6}{4} (0.35^4)(0.65^2) = 0.0951 \dots$ $P(6) + P(5) + P(4) = 0.1174$ $0.1174 < 0.12$ $\binom{6}{3} (0.35^3)(0.65^3) = 0.23549 \dots$ $P(6) + P(5) + P(4) + P(3) = 0.35289$ $0.35289 > 0.12$ $\therefore n = 3$	Scale 5D (0, 2, 3, 4, 5) <i>Low Partial Credit:</i> • Work of merit, for example, effort at calculating the probability of one possible result <i>Mid Partial Credit:</i> • Probability of two results set up <i>High Partial Credit:</i> • Probability of three results set up

10	Model Solution – 50 Marks	Marking Notes
(a) (i)	$\frac{82 - 78}{6} = 0.67$ 74.86%	Scale 10C (0, 4, 7, 10) <i>Low Partial Credit:</i> - Some correct substitution into formula <i>High Partial Credit:</i> 3 correct new entries
(a) (ii)	$\frac{70 - 78}{6} = -1.33$ $P(Z > -1.33) = 0.9082$ $0.9082 - 0.5 = 0.4082$ $P(Z > x) = 1 - 0.4082$ 0.5918 $x = 0.23$ $78 + 6(0.23) = 79.38$ $= 79.4$	Scale 10D (0, 3, 5, 7, 10) <i>Low Partial Credit:</i> - Work of merit, for example, relevant diagram drawn or substitution into formula <i>Mid Partial Credit:</i> - Probability of two results set up <i>High Partial Credit:</i> - Probability of three results set up <i>Full Credit -1</i> - Apply a * for no or incorrect rounding
(b)	Null Hypothesis: $\mu = 78$ Alternative Hypothesis: $\mu \neq 78$ Calculations: $Z = \frac{71 - 78}{\frac{6}{\sqrt{500}}} = -26.087 \dots$ Conclusion: Reject H_0 Driving times have changed Reason: $-26.087 < -1.96$	Scale 10D (0, 3, 5, 7, 10) Note: 4 steps to check: - Hypotheses - Calculations - Conclusion - Reason Reason may be included with Calculations (or Conclusion), and must be based on relevant calculations – if either a confidence interval or test statistic is not found, a conclusion cannot be drawn If the null or alternative hypothesis is incorrect, then Step 1 is not considered correct. In this case, mark Steps 2 and 4 as if the null and alternative hypothesis had been given. However, Step 3 (conclusion) cannot be considered correct, so award <i>Mid Partial Credit</i> at most. <i>Low Partial Credit:</i> - Work of merit in one step <i>Mid Partial Credit:</i> - Two steps correct <i>High Partial Credit</i> - Three steps correct

10	Model Solution – 50 Marks	Marking Notes
(c) (i)	$\binom{4}{2} \binom{6}{3}$ $= 120$	Scale 10C (0, 4, 7, 10) <i>Low Partial Credit:</i> $\binom{4}{2}$ or $\binom{6}{3}$ <i>High Partial Credit:</i> $\binom{4}{2}$ and $\binom{6}{3}$
(c) (ii)	5W $\binom{6}{5} = 6$ 4W 1M $\binom{6}{4} \binom{4}{1} = 60$ 3W 2M $\binom{6}{3} \binom{4}{2} = 120$ $= 186$	Scale 5C (0, 2, 3, 5) <i>Low Partial Credit:</i> - Work of merit, for example, one possible committee written - One possible outcome formulated <i>High Partial Credit</i> - Two possible outcomes formulated
(c) (iii)	$\binom{10}{5} = 252$ $\frac{1}{251}$	Scale 5C (0, 2, 3, 5) <i>Low Partial Credit:</i> - Work of merit, for example, effort towards finding a relevant combination <i>High Partial Credit</i> 252 and stops

